

Test III [13/11/2025]: Every answer requires proper justification for award of marks.

1. (15 points) Let $f : \{0, 1\}^2 \mapsto \{0, 1\}$ be a boolean function given by $f(ab) = a \wedge b$, where $\wedge$ is the AND operator. Let $U_f : \{0, 1\}^3 \mapsto \{0, 1\}^3$ be the quantum implementation of f with the transformation $U_f : |xyz\rangle = |xy\rangle \otimes |z \oplus f(xy)\rangle$.

a) Write down the unitary matrix corresponding to U_f .

Soln: The map is $U_f |i\rangle = |i\rangle$ for $0 \leq i \leq 5$. $U_f |6\rangle = |7\rangle$ and $U_f |7\rangle = |6\rangle$. In the resultant 8×8 boolean matrix A , $A[i, i] = 1$ for $0 \leq i \leq 5$, $A[6, 7] = A[7, 6] = 1$. All the remaining entries will be zero.

b) Argue that the matrix U_f is unitary.

Soln: The matrix A is real symmetric has the property that $A^2 = I$. Hence, $A^* A = A^T A = A^2 = I$, proving that A is unitary.

c) Let f be arbitrary (not necessarily $a \wedge b$). Show that for any $x \in \{0, 1\}^2$, if U_f is acted upon the state $|x\rangle \otimes |-\rangle$, the resultant output state is $(-1)^{f(x)} |x\rangle \otimes |-\rangle$, where $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Soln: $U_f |x0\rangle = |xf(x)\rangle$ and $U_f |x1\rangle = |x\overline{f(x)}\rangle$. Hence, $U_f \frac{1}{\sqrt{2}}(|x0\rangle - |x1\rangle) = \frac{1}{\sqrt{2}}(|xf(x)\rangle - |x\overline{f(x)}\rangle)$. If $f(x) = 0$, this is $\frac{1}{\sqrt{2}}(|x0\rangle - |x1\rangle) = |x-\rangle$. If $f(x) = 1$, this is $\frac{1}{\sqrt{2}}(|x0\rangle - |x1\rangle) = -|x-\rangle$. We can combine this into $(-1)^{f(x)} |x-\rangle$.

d) Suppose that the input to U_f is a uniform superposition of all basis states. Suppose that U_f corresponds to the function $f(ab) = a \oplus b$. Further, the first two bits of the output of U_f is acted upon by $H \otimes H$, and those two bits are measured. What is the probability that the measurement outcome is:

a) $|00\rangle$? b) $|11\rangle$?

Soln: The circuit corresponds to the Deutsch Jozsa algorithm and the coefficient of the output state $|z\rangle$ upto a normalizing constant is: $(-1)^{f(00) \oplus \langle 00, z \rangle} + (-1)^{f(01) \oplus \langle 01, z \rangle} + (-1)^{f(10) \oplus \langle 10, z \rangle} + (-1)^{f(11) \oplus \langle 11, z \rangle}$.

When $z \in \{|00\rangle, |01\rangle, |10\rangle\}$, the above expression evaluates to 0. The value is non-zero when $z = |11\rangle$. Hence, the result will always be $|11\rangle$.

2. (3 points) Consider the function $f : \{0, 1\}^2 \mapsto \{0, 1\}$ where $f(01) = 1$, and $f(ab) = 0$ if $(ab) \neq (01)$. In the quantum circuit of the Grover's algorithm, recall that after performing $U_f ((H \otimes H) |00\rangle) \otimes |-\rangle$, the first two bits of the result is acted upon by the inversion operator $I - 2aa^T$, where a is the unit vector along the direction $\vec{1}$. This completes one round of Grover's iteration. At this time, what will be the resultant state of the system?

Soln: After $U_f ((H \otimes H) |00\rangle)$, the state vector will have the form $\frac{1}{2}[1, -1, 1, 1]$. After application of the diffusion operator, the state will be $[0, 1, 0, 0]$, which corresponds to the state $|01\rangle$, which is the required solution.

3. (3 points) The Grover's algorithm prepares the uniform state $\frac{1}{\sqrt{2^n}} \sum_i |i\rangle \otimes |-\rangle$ (upto a normalizing constant) on an n input qubit system; first applies U_f ; and then the inversion around the mean operator. Suppose that f is defined by $f(x) = 1$ when $x = x_0$ and $f(x) = 0$. Suppose the resultant state is $-\alpha |x_0\rangle + \sum_{x \neq x_0} \beta |x\rangle$, where α, β are real numbers. Show that in the case of $n = 16$, $\frac{1}{4} \leq \alpha \leq \frac{1}{2}$.

Soln: The details of the required proof is posted in the notes (see Proposition 2.4 of Grover's algorithm notes). Here, $\alpha_0 = \frac{1}{\sqrt{2^{16}}}$ and $\alpha_1 \leq \alpha_0 + \frac{2}{\sqrt{2^{16}}} \leq \frac{3}{256}$. The lower bound can't be obtained with $n = 16$.

4. (6 points) Consider the two bit function $f : \{0, 1\}^2 \mapsto \{0, 1\}^2$. Suppose that $f(x) = f(y)$ if and only if $x \oplus y = 10$. Let U_f be the corresponding quantum gate implementing f (Recall how U_f was defined in the previous questions.) Assume that U_f is applied to the state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \otimes |00\rangle$. The output of U_f is acted upon by $H \oplus H \oplus I \oplus I$, where I is the 2×2 identity matrix. What are the possible states in the output of the circuit? (That is, what basis states have non-zero probability of occurrence?). Give clear justification for your answer. (Hint: This is a standard circuit, done in the class.)

Soln: The given circuit implements the Simon's algorithm. For each $x, z \in \{0, 1\}^2$, the coefficient of $|z\rangle \otimes |f(x)\rangle$ is $\sum_x \sum_z (-1)^{\langle x, z \rangle}$. Moreover, non-zero probability is possible only if $\langle x, 10 \rangle = 0$. This limits the possible $|x\rangle$ values to $|01\rangle$ and $|00\rangle$. Moreover, $f(00) = f(10)$, $f(01) = f(11)$. Hence, non-zero amplitudes are possible only form $|00\rangle \otimes |f(00)\rangle$, $|00\rangle \otimes |f(01)\rangle$, $|01\rangle \otimes |f(00)\rangle$, $|01\rangle \otimes |f(01)\rangle$.